Bitwise Operators - Zero to Hero Questions List

May 26, 2026

1. Write a C program to find the bitwise AND of two numbers. *(Hint: Use `&` operator)*
2. Write a C program to find the bitwise OR of two numbers. *(Hint: Use `|` operator)*
3. Write a C program to find the bitwise XOR of two numbers. *(Hint: Use `^` operator)*
4. Write a C program to find the bitwise complement of a number. *(Hint: Use `~` operator)*
5. Write a C program to left shift a number by 1 position. *(Hint: Use `<<` operator)*
6. Write a C program to right shift a number by 1 position. *(Hint: Use `>>` operator)*
7. Write a program to check whether a number is even or odd using bitwise operator. *(Hint: Check last bit using `& 1`)*
8. Write a program to swap two numbers using XOR. *(Hint: Use three XOR operations)*
9. Write a program to check whether the ith bit is set or not. *(Hint: Use `1 << i`)*
10. Write a program to set the ith bit of a number. *(Hint: Use OR with mask)*
11. Write a program to clear the ith bit of a number. *(Hint: Use AND with NOT mask)*
12. Write a program to toggle the ith bit of a number. *(Hint: Use XOR with mask)*
13. Write a program to count the number of set bits in a number. *(Hint: Repeatedly use `n & 1`)*
14. Write a program to check whether a number is a power of 2. *(Hint: Use `n & (n-1)`)*
15. Write a program to find the position of the first set bit. *(Hint: Right shift until bit becomes 1)*
16. Write a program to multiply a number by 2 using bitwise operator. *(Hint: Left shift by 1)*
17. Write a program to divide a number by 2 using bitwise operator. *(Hint: Right shift by 1)*
18. Write a program to convert lowercase alphabet to uppercase using bitwise operator. *(Hint: ASCII manipulation)*
19. Write a program to convert uppercase alphabet to lowercase using bitwise operator. *(Hint: Use bit setting)*
20. Write a program to check whether two numbers are equal using XOR. *(Hint: XOR result becomes 0)*
21. Write a program to find the only non-repeating element in an array where every other element repeats twice. *(Hint: XOR all elements)*
22. Write a program to reverse the bits of a number. *(Hint: Extract bits one by one)*

23. Write a program to count leading zeros in binary representation. *(Hint: Check bits from MSB side)*

24. Write a program to count trailing zeros in binary representation. *(Hint: Check bits from LSB side)*

25. Write a program to check whether a number is positive or negative using bitwise operator. *(Hint: Check sign bit)*

26. Write a program to turn off the rightmost set bit. *(Hint: Use `n & (n-1)`)*

27. Write a program to isolate the rightmost set bit. *(Hint: Use `n & -n`)*

28. Write a program to set all bits below the ith bit. *(Hint: Create mask using shifts)*

29. Write a program to check if two integers have opposite signs. *(Hint: XOR sign bits)*

30. Write a program to rotate bits of a number to the left. *(Hint: Combine shifts)*

31. Write a program to rotate bits of a number to the right. *(Hint: Use circular shifting)*

32. Write a program to count total bits required to represent a number in binary. *(Hint: Keep right shifting)*

33. Write a program to print binary representation of a number. *(Hint: Use bit masking)*

34. Write a program to find whether the kth bit from right is 1 or 0. *(Hint: Shift and AND)*

35. Write a program to clear all bits from LSB to ith bit. *(Hint: Use left-shifted mask)*

36. Write a program to clear all bits from MSB to ith bit. *(Hint: Use mask creation)*

37. Write a program to copy set bits from one number to another. *(Hint: Use AND and OR)*

38. Write a program to compare two integers without using comparison operators. *(Hint: Use XOR and shifts)*

39. Write a program to add two numbers without using `+` operator. *(Hint: Use XOR and AND)*

40. Write a program to subtract two numbers without using `-` operator. *(Hint: Use bitwise borrow logic)*

41. Write a program to find maximum of two numbers using bitwise operators. *(Hint: Use sign difference)*

42. Write a program to find minimum of two numbers using bitwise operators. *(Hint: Similar to max logic)*

43. Write a program to toggle all bits of a number except MSB. *(Hint: Create suitable mask)*

44. Write a program to check whether binary representation is palindrome or not. *(Hint: Reverse bits and compare)*

45. Write a program to find parity of a number (odd/even set bits). *(Hint: XOR all bits)*

46. Write a program to extract last 4 bits of a number. *(Hint: Use mask `0xF`)*

47. Write a program to merge two nibbles into one byte. *(Hint: Use shifts and OR)*

48. Write a program to exchange odd and even bits in a number. *(Hint: Use masks `0xAA` and `0x55`)*
49. Write a program to check whether a bit pattern is alternating. *(Hint: XOR adjacent bits)*
50. Write a program to implement a simple calculator using bitwise operations only. *(Hint: Use custom add/subtract logic)*